

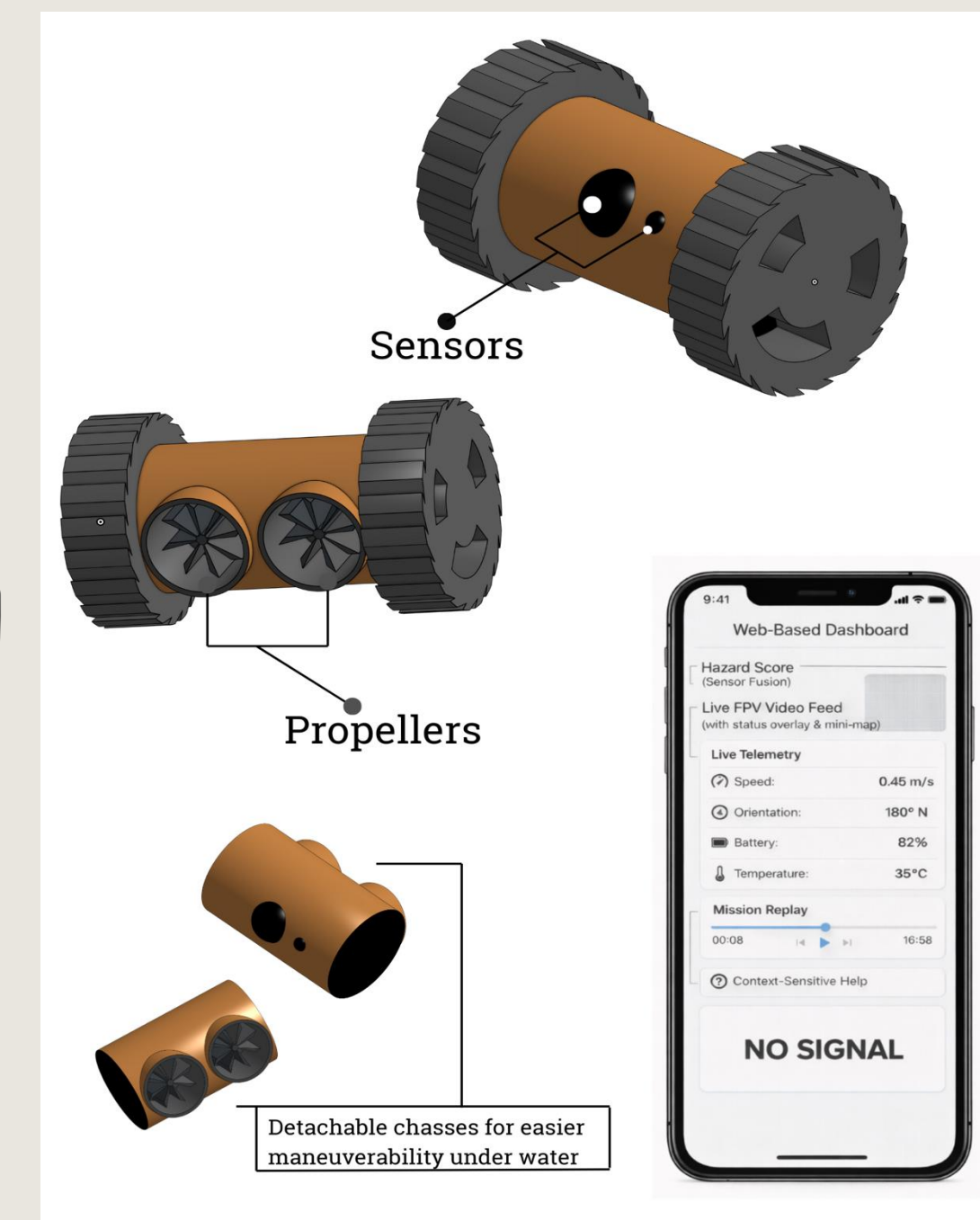
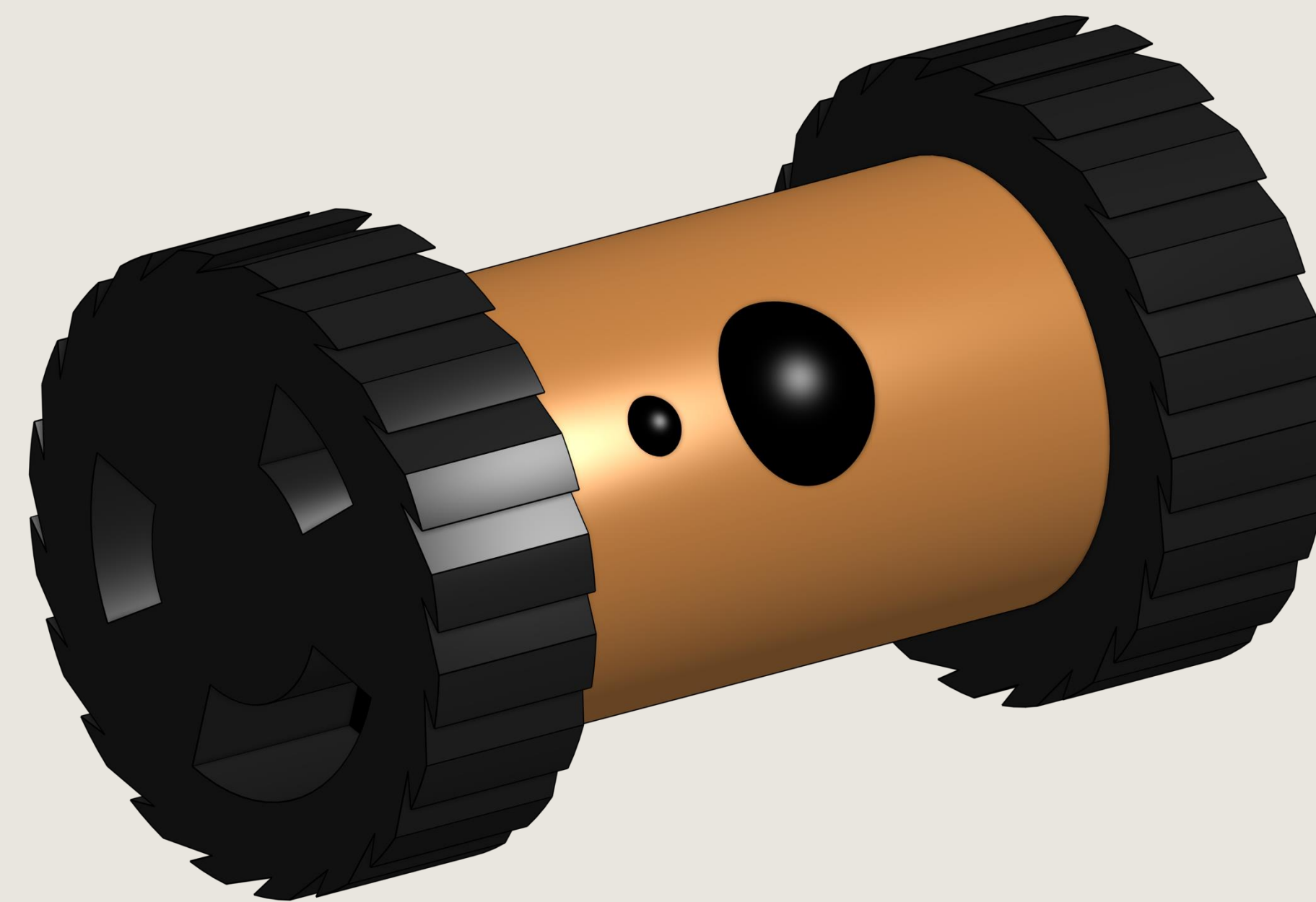
Amphibot

Embedded Systems Design Project II

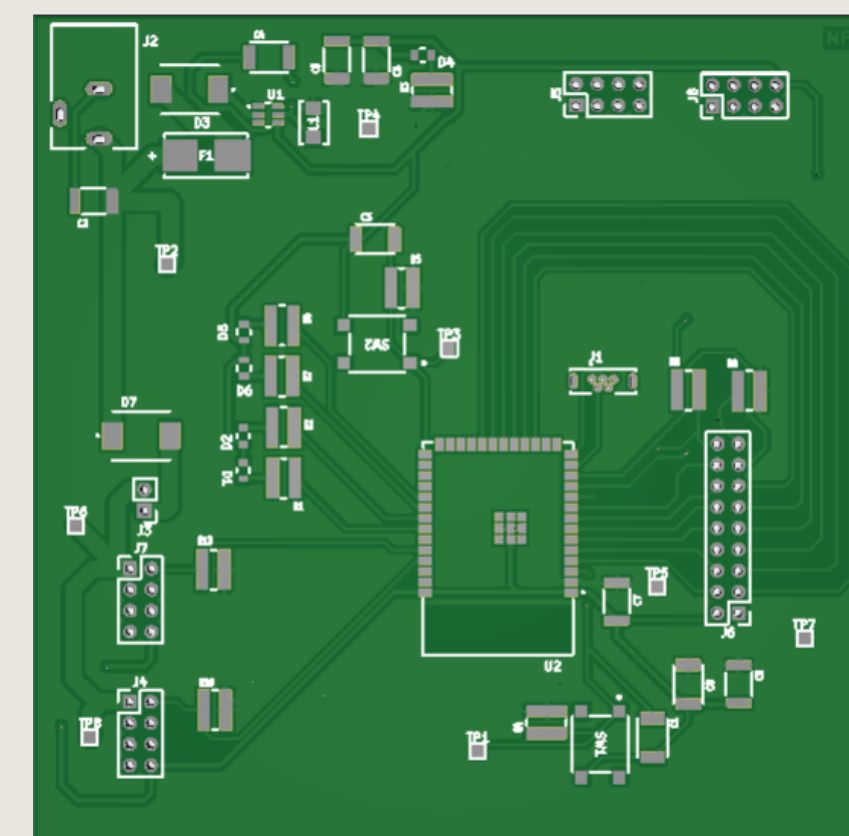
EGR314 Spring 2025

Mihir Patel, Raunak Singh, Lakshanand Sugumar

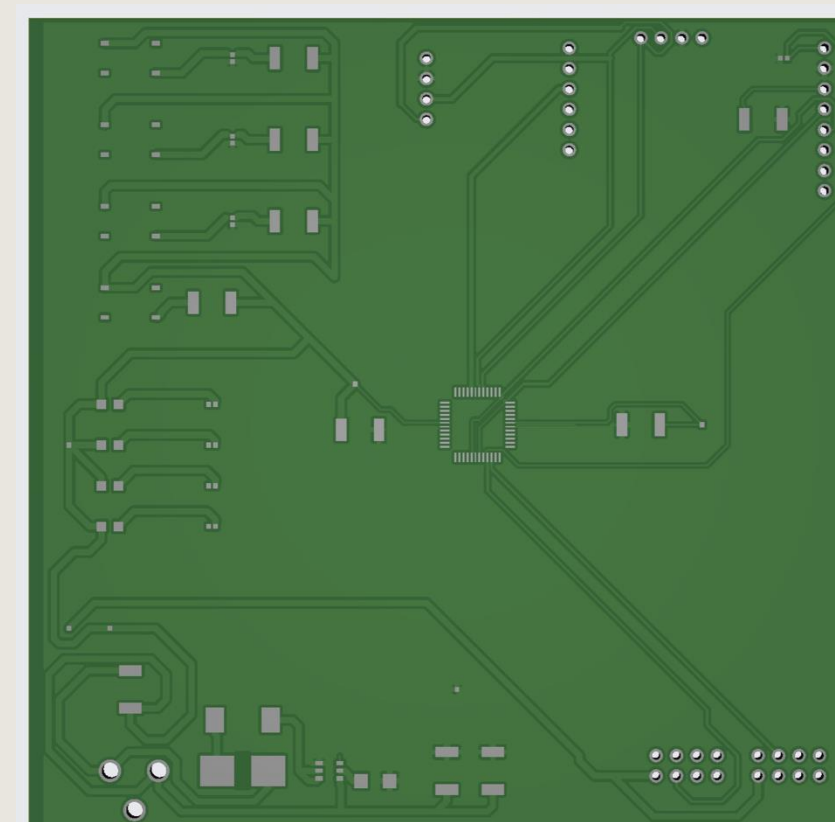
The Wi-Fi module serves as the central communication node, managing bidirectional wireless data transmission to and from the MQTT broker over Wi-Fi. The human-machine interface (HMI) PIC18F47Q10 handles sensor acquisition, interfacing with an MPU6050 IMU over I²C and presenting live readings on an OLED display for in-person operators. The PIC18F57Q83 on the actuator board receives motor speed commands over UART and generates PWM signals through an IFX9201SGAUMA1CT motor driver, providing precise closed-loop control of the robot's propulsion system.



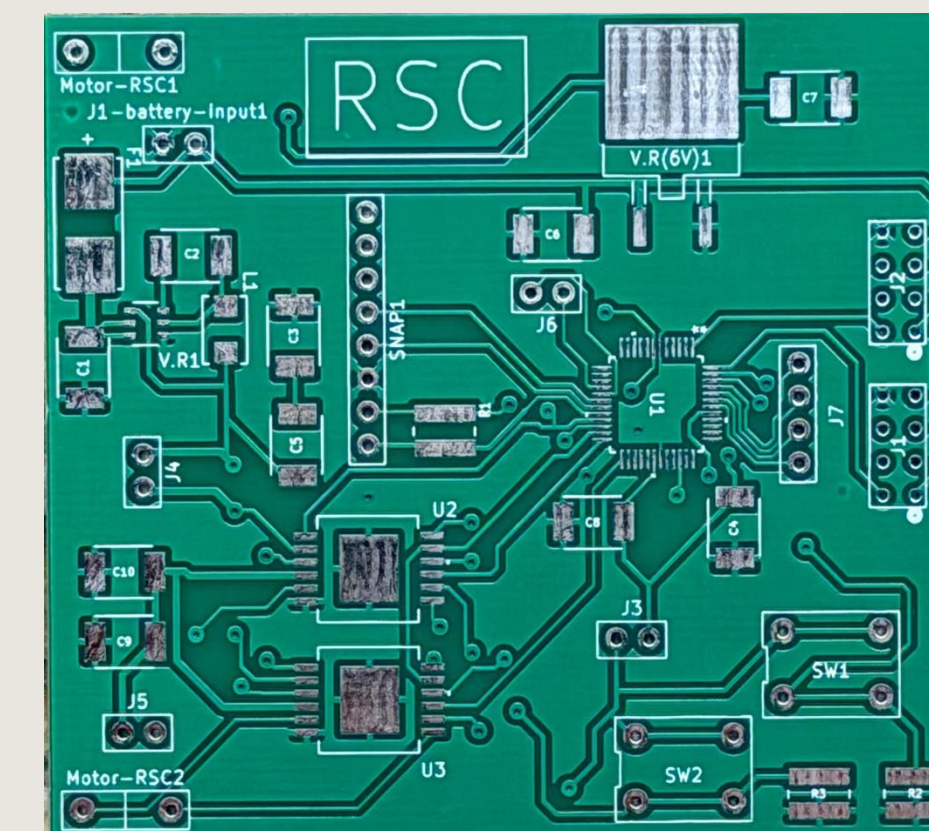
Team concept drawing of an autonomous underwater inspection robot with a detachable chassis and a dual-propeller drive system, optimized for navigation in confined aquatic environments. The robot integrates onboard inertial sensors and streams real-time telemetry to a web-based dashboard for remote monitoring. This prototype serves as a proof-of-concept for a scalable, modular underwater sensing platform.



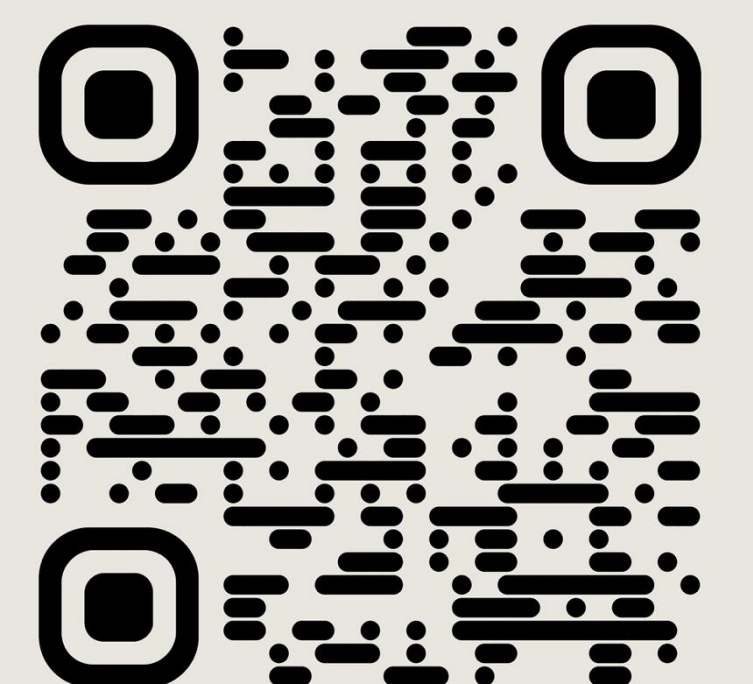
MQTT + WiFi



Sensor + HMI



Actuator



Team Website